

Algebra

CHAPTER 10

Mathematical Reasoning

10.1–10.8

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INTRODUCTION

The meaning of 'Logic' is the 'science of reasoning'. The study of logic through the use of mathematical symbols is called mathematical logic. Since the symbols are abstract and neutral, they give clear expression to our thoughts. The mathematical approach to logic was first propounded by British mathematician George Boole. On this account, the mathematical logic is also called Boolean logic.

Statement

A sentence is called a statement if it is either true or false but not both.

Statements are denoted by letters p, q, r , etc.

For example,

- (i) ' $2 \times 5 = 10$ ' is a statement because it is true.
- (ii) 'Mumbai is in America' is statement because it is false.
- (iii) 'Today is Monday' is not a statement because it is neither true nor false.

Truth value of a statement

We know that a statement is either true or false. The truth or falsity of a statement is called its truth value. If a statement is true, then its truth value is denoted by 'T' and if a statement is false then its truth value is denoted by 'F'.

For example,

- (i) The truth value of the statement '18 is prime number' is F, because this statement is false.
- (ii) The truth value of the statement '36 is divisible by 9' is T, because this statement is true.

Truth table

A table indicating the truth values of one or more statements is called a truth table.

The truth table for one statement ' p ' is:

p
T
F

The truth table for two statements p and q is:

p	q
T	T
T	F
F	T
F	F

The truth table for three statements p, q and r is:

p	q	r
T	T	T
T	T	F
T	F	T
T	F	F
F	T	T
F	T	F
F	F	T
F	F	F

In case of n statements, there are 2^n distinct possible arrangements of truth values of the statements.

Negation operation

If p is any statement, then the denial of statement p is called the negation of statement p and is written as $\sim p$. The negation of statement p is formed by inserting the word 'not' in p or by writing 'It is false that ...' before p .

For example, for the statement,

p : Sum of two odd integers is even.

$\sim p$: Sum of two odd integers is not even.

or $\sim p$: It is false that sum of two odd integers is even.

The truth value of negation of a statement is always opposite to the truth value of the original statement.

p	$\sim p$
T	F
F	T

COMPOUND STATEMENT AND BASIC LOGICAL CONNECTIVES

A statement whose truth value does not explicitly depend on another statement is called a simple statement.

For example, 'the letter e is a vowel' is a simple statement. If two or more simple statements are combined by the words 'and', 'or', 'if ...then', 'if and only if', then the resulting statement is called a compound statement. Simple statements which, on combining, form a compound statement are called component statements of the compound statement under consideration. The words 'and', 'or', 'if ...then', 'if and only if' are called logical connectives. The compound statement S consisting of component statements $p, q, r, \dots$ is written as $S(p, q, r, \dots)$.

Few compound statements are:

- (i) Meena studies in class X and resides in Delhi.

- (ii) Namrata might be watching movie or playing chess.
- (iii) If the apple is red then it is ripe.
- (iv) India can win the match if and only if Rohit hits a six on each of the remaining 3 balls.

The logical connectives used in the above compound statements and their symbols are tabulated as follows:

Basic logical connective	Symbol	Compound statement
And	$\wedge$	Conjunction
Or	$\vee$	Disjunction
If ... then	$\rightarrow$	Conditional statement
If and only if	$\leftrightarrow$	Biconditional statement

If p and q are two component statements of a compound statement, then mathematically,

- ' p and q ' can be written as ' $p \wedge q$ '.
- ' p or q ' can be written as ' $p \vee q$ '.
- 'If p then q ' can be written as ' $p \rightarrow q$ '.
- ' p if and only if q ' can be written as ' $p \leftrightarrow q$ '.

TRUTH TABLE OF COMPOUND STATEMENTS USING CONNECTIVES 'AND' AND 'OR'

Truth values of statements ' p and q ' and ' p or q ' are tabulated as follows:

p	q	$p \wedge q$	$p \vee q$
T	T	T	T
T	F	F	T
F	T	F	T
F	F	F	F

Clearly, statement ' p and q ' is true when both p and q are true. Also, this statement is false if at least one of the statements is false.

Statement ' p or q ' is true if at least one of the statements is true. Also, this statement is false if both the statements are false.

Now, we can make the truth table for the negations of these statements as follows:

p	q	$\sim(p \wedge q)$	$\sim(p \vee q)$
T	T	F	F
T	F	T	F
F	T	T	F
F	F	T	T

ILLUSTRATION 10.1

Form the truth table of $\sim p \wedge q$ and $p \vee \sim q$.

Sol.

p	q	$\sim p$	$\sim q$	$\sim p \wedge q$	$p \vee \sim q$
T	T	F	F	F	T
T	F	F	T	F	T
F	T	T	F	T	F
F	F	T	T	F	T

ILLUSTRATION 10.2

Find the truth values of
(i) $\sim(p \vee \sim q)$ (ii) $\sim(\sim p \wedge \sim q)$

Sol.

(i) $\sim(p \vee \sim q)$

p	q	$\sim q$	$p \vee \sim q$	$\sim(p \vee \sim q)$
T	T	F	T	F
T	F	T	T	F
F	T	F	F	T
F	F	T	T	F

(ii) $\sim(\sim p \wedge \sim q)$

p	q	$\sim p$	$\sim q$	$\sim p \wedge \sim q$	$\sim(\sim p \wedge \sim q)$
T	T	F	F	F	T
T	F	F	T	F	T
F	T	T	F	F	T
F	F	T	T	T	F

ILLUSTRATION 10.3

Find the truth values of the following compound statements:
(i) $p \wedge (q \wedge r)$ (ii) $(p \vee q) \vee r$
(iii) $p \wedge (q \vee r)$ (iv) $(p \wedge q) \vee r$

Sol.

p	q	r	$p \wedge (q \wedge r)$	$(p \vee q) \vee r$	$p \wedge (q \vee r)$	$(p \wedge q) \vee r$
T	T	T	T	T	T	T
T	T	F	F	T	T	T
T	F	T	F	T	T	T
F	T	T	F	T	F	T
T	F	F	F	T	F	F
F	T	F	F	T	F	F
F	F	T	F	T	F	T
F	F	F	F	F	F	F

CONDITIONAL STATEMENTS

Consider the following compound statements:
If Bhopal is in India then $2 + 7 = 9$.
If Bhopal is in India then $2 + 7 = 10$.
If Bhopal is in England then $2 + 7 = 9$.
If Bhopal is in England then $2 + 7 = 10$.
If first statement is p and second statement is q , then we have following truth table for $p \rightarrow q$.

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

ILLUSTRATION 10.4

Find the truth values of

- (i)
- $\sim p \rightarrow q$
- (ii)
- $\sim (p \rightarrow q)$

Sol.

- (i)
- $\sim p \rightarrow q$

p	q	$\sim p$	$\sim p \rightarrow q$
T	T	F	T
T	F	F	T
F	T	T	T
F	F	T	F

- (ii)
- $\sim (p \rightarrow q)$

p	q	$p \rightarrow q$	$\sim (p \rightarrow q)$
T	T	T	F
T	F	F	T
F	T	T	F
F	F	T	F

BICONDITIONAL STATEMENTS

Consider the following compound statements:

Bhopal is in India if and only if $2 + 7 = 9$.**Bhopal is in India** if and only if $2 + 7 = 10$.**Bhopal is in England** if and only if $2 + 7 = 9$.**Bhopal is in England** if and only if $2 + 7 = 10$.If first statement is p and second statement is q , then we have following truth table for $p \leftrightarrow q$:

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

ILLUSTRATION 10.5

Find the truth values of

- (i)
- $\sim p \leftrightarrow q$
- (ii)
- $\sim (p \leftrightarrow q)$

Sol.

- (i)
- $\sim p \leftrightarrow q$

p	q	$\sim p$	$\sim p \leftrightarrow q$
T	T	F	F
T	F	F	T
F	T	T	T
F	F	T	F

- (ii)
- $\sim (p \leftrightarrow q)$

p	q	$p \leftrightarrow q$	$\sim (p \leftrightarrow q)$
T	T	T	F
T	F	F	T
F	T	F	T
F	F	T	F

LOGICAL EQUIVALENCE

Two compound statements $S_1(p, q, r, \dots)$ and $S_2(p, q, r, \dots)$ of component statements $p, q, r, \dots$ are called logically equivalent or simply equivalent or equal if they have identical truth values and we write $S_1(p, q, r, \dots) \equiv S_2(p, q, r, \dots)$.

ILLUSTRATION 10.6Show that the compound statements $(p \vee q) \wedge \sim p$ and $\sim p \wedge q$ are logically equivalent.**Sol.** Truth values of $(p \vee q) \wedge \sim p$ and $\sim p \wedge q$ are as follows:

p	q	$\sim p$	$p \vee q$	$(p \vee q) \wedge \sim p$	$\sim p \wedge q$
T	T	F	T	F	F
T	F	F	T	F	F
F	T	T	T	T	T
F	F	T	F	F	F

Clearly, $(p \vee q) \wedge \sim p$ and $\sim p \wedge q$ have identical truth values. $\therefore (p \vee q) \wedge \sim p \equiv \sim p \wedge q$.**ILLUSTRATION 10.7**Show that $(p \vee q) \rightarrow r \equiv (p \rightarrow r) \wedge (q \rightarrow r)$.**Sol.** Truth table for $(p \vee q) \rightarrow r$ and $(p \rightarrow r) \wedge (q \rightarrow r)$.

p	q	r	$p \vee q$	$p \rightarrow r$	$q \rightarrow r$	$(p \vee q) \rightarrow r$	$(p \rightarrow r) \wedge (q \rightarrow r)$
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	F
T	F	T	T	T	T	T	T
T	F	F	T	F	T	F	F
F	T	T	T	T	T	T	T
F	T	F	T	T	F	F	F
F	F	T	F	T	T	T	T
F	F	F	F	T	T	T	T

It is evident from the last two columns that

$$(p \vee q) \rightarrow r \equiv (p \rightarrow r) \wedge (q \rightarrow r).$$

Note:

- $p \rightarrow q \equiv (\sim p) \vee q$
- $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$
- $p \leftrightarrow q \equiv (\sim p \vee q) \wedge (p \vee \sim q)$

CONVERSE, INVERSE AND CONTRAPOSITIVE OF STATEMENTGiven an if-then statement “if p , then q ” we can create three related statements.

A conditional statement consists of two parts, a hypothesis in the “if” clause and a conclusion in the “then” clause.

For instance, consider the compound statement,

“If it rains, then they cancel the school.” ... (1)

Here,

“It rains” is the hypothesis.

“They cancel the school” is the conclusion.

To form the converse of the conditional statement, just interchange the hypothesis and the conclusion. The converse of statement (1) is “If they cancel the school, then it rains.”

To form the inverse of the conditional statement, take the negation of both the hypothesis and the conclusion. The inverse of statement (1) is “If it does not rain, then they do not cancel the school.”

To form the contrapositive of the conditional statement, interchange the hypothesis and the conclusion of the inverse statement. The contrapositive of statement (1) is “If they do not cancel the school, then it does not rain.”

Statement	If p then q ($p \rightarrow q$)
Converse	If q then p ($q \rightarrow p$)
Inverse	If not p then not q ($\sim p \rightarrow \sim q$)
Contrapositive	If not q then not p ($\sim q \rightarrow \sim p$)

If the statement is true, then the contrapositive is also logically true. If the converse is true, then the inverse is also logically true.

ILLUSTRATION 10.8

For the statement “If two angles are congruent, then they have the same measure”, write the converse, inverse and contrapositive statements.

Sol. Converse: If two angles have the same measure, then they are congruent.

Inverse: If two angles are not congruent, then they do not have the same measure.

Contrapositive: If two angles do not have the same measure, then they are not congruent.

In the above example, since the hypothesis and conclusion are equivalent, all four statements are true. But this will not always be the case. Consider the last question in the following concept application exercise.

CONCEPT APPLICATION EXERCISE 10.1

1. Write down the truth table for the compound statement: $(\sim p \vee q) \wedge (\sim p \wedge \sim q)$.
2. Find the truth values of the following compound statements:
(a) $(p \vee \sim r) \wedge (q \vee \sim r)$ (b) $\sim (p \vee \sim q) \wedge (\sim p \vee r)$.
3. Find the truth values of
(a) $\sim p \rightarrow (q \rightarrow p)$ (b) $(p \rightarrow q) \rightarrow (p \wedge q)$.
4. Find the truth values of
(a) $(p \leftrightarrow \sim q) \leftrightarrow (q \rightarrow p)$ (b) $(p \rightarrow q) \vee \sim (p \leftrightarrow \sim q)$
5. Construct the truth tables for the following statements:
(a) $(p \wedge q) \rightarrow \sim P$ (b) $(p \wedge q) \rightarrow (p \vee q)$
(c) $(p \wedge q) \rightarrow r$ (d) $[p \wedge (\sim r)] \rightarrow (q \vee r)$
6. Show that $\sim (p \leftrightarrow q) \equiv (p \wedge \sim q) \vee (\sim p \wedge q)$.
7. Are the following statements equivalent:
‘If the trades do not reduce the price then the government will take action against them’. ‘It is not true that the traders do not reduce the prices and government does not take action against them’.
8. For the statement: “If a quadrilateral is a rectangle, then it has two pairs of parallel sides”, write the converse, inverse and contrapositive statements.

TAUTOLOGY AND CONTRADICTION (FALLACY)

Tautology: A compound statement is called a tautology if it is always true for all possible truth values of its component statements.

Contradiction: A compound statement is called a contradiction if it is always false for all possible truth values of its component statements. A contradiction is also called a **fallacy**.

The negation of a tautology is a contradiction and negation of a contradiction is a tautology.

ILLUSTRATION 10.9

- Show that
- (i) $p \rightarrow (p \vee q)$ is a tautology
- (ii) $(p \vee q) \wedge (\sim p \wedge \sim q)$ is a contradiction

Sol.

- (i) Truth values of $p \rightarrow (p \vee q)$

p	q	$p \vee q$	$p \rightarrow (p \vee q)$
T	T	T	T
T	F	T	T
F	T	T	T
F	F	F	T

Thus, for all possible truth values of p and q , the compound statement $p \rightarrow (p \vee q)$ is true.

Hence, $p \rightarrow (p \vee q)$ is a tautology.

- (ii) Truth values of $(p \vee q) \wedge (\sim p \wedge \sim q)$

p	q	$\sim p$	$\sim q$	$p \vee q$	$\sim p \wedge \sim q$	$(p \vee q) \wedge (\sim p \wedge \sim q)$
T	T	F	F	T	F	F
T	F	F	T	T	F	F
F	T	T	F	T	F	F
F	F	T	T	F	T	F

Thus, for all possible truth values of p and q , the compound statement $(p \vee q) \wedge (\sim p \wedge \sim q)$ is false.

Hence, $(p \vee q) \wedge (\sim p \wedge \sim q)$ is a contradiction.

ILLUSTRATION 10.10

Show that $[(p \vee q) \vee r] \leftrightarrow [p \vee (q \vee r)]$ is a tautology.

Sol. Truth value of $[(p \vee q) \vee r] \leftrightarrow [p \vee (q \vee r)]$

p	q	r	$p \vee q$	$(p \vee q) \vee r$	$q \vee r$	$p \vee (q \vee r)$	$[(p \vee q) \vee r] \leftrightarrow [p \vee (q \vee r)]$
T	T	T	T	T	T	T	T
T	T	F	T	T	T	T	T
T	F	T	T	T	T	T	T
F	T	T	T	T	T	T	T
T	F	F	T	T	F	T	T
F	T	F	T	T	T	T	T
F	F	T	F	T	T	T	T
F	F	F	F	F	F	F	T

Thus, for all possible truth values of p , q and r , the compound statement: $[p \vee q] \vee r \leftrightarrow [p \vee (q \vee r)]$ is true.

$\therefore [(p \vee q) \vee r] \leftrightarrow [p \vee (q \vee r)]$ is a tautology.

ALGEBRA OF STATEMENTS

Idempotent Laws: If p is any statement, then

$$(i) p \vee p \equiv p \quad (ii) p \wedge p \equiv p$$

Proof:

Truth values $p \vee p$ and $p \wedge p$

p	$p \vee p$	$p \wedge p$
T	T	T
F	F	F

$\therefore p \vee p \equiv p$ and $p \wedge p \equiv p$

Complement Laws: If p is any statement, then

$$(i) p \vee \sim p \equiv t \quad (ii) p \wedge \sim p \equiv f \quad (iii) \sim \sim p \equiv p$$

$$(iv) \sim t \equiv f \quad (v) \sim f \equiv t$$

Here, t and f are, respectively, some tautology and fallacy in terms of the statement p .

Proof:

Truth values of $p \vee \sim p$, $p \wedge \sim p$, $\sim \sim p$, $\sim t$, $\sim f$

p	$\sim p$	$p \vee \sim p$	t	$p \wedge \sim p$	f	$\sim \sim p$	$\sim t$	$\sim f$
T	F	T	T	F	F	T	F	T
F	T	T	T	F	F	F	F	T

$\therefore p \vee \sim p \equiv t$, $p \wedge \sim p \equiv f$, $\sim(\sim p) \equiv p$, $\sim t \equiv f$, $\sim f \equiv t$

Identity Laws: If p is any statement, then

$$(i) p \wedge t \equiv p \quad (ii) p \wedge f \equiv f$$

$$(iii) p \vee t \equiv t \quad (iv) p \vee f \equiv p$$

Here, t and f are, respectively, some tautology and fallacy in terms of the statement p .

Proof:

Truth values of $p \wedge t$, $p \wedge f$, $p \vee t$, $p \vee f$

p	t	f	$p \wedge t$	$p \wedge f$	$p \vee t$	$p \vee f$
T	T	F	T	F	T	T
F	T	F	F	F	T	F

$\therefore p \wedge t \equiv p$, $p \wedge f \equiv f$, $p \vee t \equiv t$, $p \vee f \equiv p$

Commutative Laws: If p and q be any two statements then

$$(i) p \wedge q \equiv q \wedge p \quad (ii) p \vee q \equiv q \vee p$$

De Morgan's Laws: If p and q be any two statements, then

$$(i) \sim(p \vee q) \equiv \sim p \wedge \sim q \quad (ii) \sim(p \wedge q) \equiv \sim p \vee \sim q$$

Proof:

(i) Truth values of $\sim(p \vee q)$ and $\sim p \wedge \sim q$ are as follows:

p	q	$\sim p$	$\sim q$	$p \vee q$	$\sim(p \vee q)$	$\sim p \wedge \sim q$
T	T	F	F	T	F	F
T	F	F	T	T	F	F
F	T	T	F	T	F	F
F	F	T	T	F	T	T

Thus, $\sim(p \vee q)$ and $\sim p \wedge \sim q$ have identical truth values.

$$\therefore \sim(p \vee q) \equiv \sim p \wedge \sim q$$

(ii) Truth values of $\sim(p \wedge q)$ and $\sim p \vee \sim q$ are as follows:

p	q	$\sim p$	$\sim q$	$p \wedge q$	$\sim(p \wedge q)$	$\sim p \vee \sim q$
T	T	F	F	T	F	F
T	F	F	T	F	T	T
F	T	T	F	F	T	T
F	F	T	T	F	T	T

Thus, $\sim(p \wedge q)$ and $\sim p \vee \sim q$ have identical truth values.

$$\therefore \sim(p \wedge q) \equiv \sim p \vee \sim q$$

Associative Laws: If p , q and r are any three statements, then

$$(i) (p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$$

$$(ii) (p \vee q) \vee r \equiv p \vee (q \vee r)$$

Distributive Laws: If p , q and r are any three statements, then

$$(i) p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$$

$$(ii) p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r).$$

ILLUSTRATION 10.11

Write the negation of statement “ $2 + 3 = 5$ and $8 < 10$ ”.

Sol. Let $p : 2 + 3 = 5$; $q : 8 < 10$

Given proposition is: $p \wedge q$

Its negation is $\sim(p \wedge q) \equiv \sim p \vee \sim q$

Therefore, the required negation is “ $2 + 3 \neq 5$ or $8 \not< 10$ ”.

ILLUSTRATION 10.12

Prove that $q \wedge \sim p \equiv \sim(q \rightarrow p)$.

Sol. $\sim(q \rightarrow p) \equiv \sim(\sim q \vee p)$
 $\equiv \sim(\sim q) \wedge \sim p$
 $\equiv q \wedge \sim p$

ILLUSTRATION 10.13

Write the negation of the compound proposition “If the examination is difficult, then I shall pass if I study hard”.

Sol. p : examination is difficult,
 q : I shall pass,
 r : I study hard

Given result is : $p \rightarrow (r \rightarrow q)$

Now, $\sim(r \rightarrow q) \equiv r \wedge \sim q$

$$\therefore \sim(p \rightarrow (r \rightarrow q)) \equiv p \wedge (r \wedge \sim q)$$

Therefore, required negation is, “The examination is difficult and I study hard but I shall not pass”

CONCEPT APPLICATION EXERCISE 10.2

1. Show that $(p \wedge q) \vee (\sim p) \vee (p \wedge \sim q)$ is a tautology.
2. Show that $[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$ is a tautology.
3. Prove that $\sim((\sim p) \wedge q) \equiv p \vee (\sim q)$.
4. Prove that $(p \wedge q) \wedge \sim(p \vee q)$ is a contradiction.
5. Prove that $\sim(\sim p \rightarrow \sim q) \equiv \sim p \wedge q$.

Exercises

Single Correct Answer Type

- Which of the following is not a statement?
 - 2 is an odd number
 - 10 is less than 8
 - the number 13 is prime
 - please do me a favour
- If p : 'Ram is tall' and q : 'Ram is intelligent', then the statement $\sim p \vee q$ is
 - Ram is not tall or he is intelligent.
 - Ram is tall or he is intelligent.
 - Ram is not tall and he is intelligent.
 - Ram is not tall then he is intelligent.
- Consider the statement p : 'New Delhi is a city'. Which of the following is not negation of p ?
 - New Delhi is not a city
 - It is false that New Delhi is a city
 - It is not the case that New Delhi is a city
 - None of these
- Which of the following is the inverse of the proposition 'If a number is a prime then it is odd'?
 - If a number is not odd then it is not a prime.
 - If a number is a prime then it is odd.
 - If a number is not odd then it is a prime.
 - If a number is not a prime then it is not odd.
- Consider the following statements:
 p : It rains today
 q : I go to school
 r : I shall meet any friends
 s : I shall go for a movie
 Then which of the following propositions represents 'If it does not rain or if I do not go to school, then I shall meet my friend and go for a movie.'?
 - $\sim(p \wedge q) \rightarrow (r \wedge s)$
 - $\sim(p \wedge \sim q) \rightarrow (r \wedge s)$
 - $\sim(p \wedge q) \rightarrow (r \vee s)$
 - None of these
- Negation of 'Paris is in France and London is in England' is
 - Paris is in England and London is in France
 - Paris is not in France or London is not in England
 - Paris is in England or London is in France
 - None of these
- Which of the following is not a proposition?
 - $\sqrt{3}$ is a prime
 - $\sqrt{2}$ is irrational
 - Mathematics is interesting
 - 5 is an even integer
- $\sim((\sim(\sim p)) \wedge q)$ is equal to
 - $\sim p \wedge q$
 - $\sim p \vee \sim q$
 - $p \wedge \sim q$
 - $\sim p \wedge \sim q$
- $\sim(p \vee (\sim q))$ is equal to
 - $\sim p \vee q$
 - $(\sim p) \wedge q$
 - $\sim p \vee \sim p$
 - $\sim p \wedge \sim q$
- Which of the following is logically equivalent to $\sim(\sim p \rightarrow q)$?
 - $p \wedge q$
 - $p \wedge \sim q$
 - $\sim p \wedge q$
 - $\sim p \wedge \sim q$
- If p , q and r are simple propositions with truth values T, F and T, respectively, then the truth value of $(\sim p \vee q) \wedge \sim r \rightarrow p$ is
 - True
 - False
 - True if r is false
 - True if q is true
- If the statements $(p \wedge \sim r) \rightarrow (q \vee r)$, q and r are all false, then p
 - is true
 - is false
 - may be true or false
 - data is insufficient
- If p , q and r are simple propositions such that $(p \wedge q) \wedge (q \wedge r)$ is true, then
 - p , q and r are all false
 - p , q and r are all true
 - p , q are true and r is false
 - p is true and q , r are false
- $\sim(p \vee (\sim p \vee q))$ is equal to
 - $\sim p \wedge (p \wedge \sim q)$
 - $(p \wedge \sim q) \vee \sim p$
 - $(p \vee \sim q) \vee \sim p$
 - None of these
- $(\sim(p \vee q)) \vee (\sim p \wedge q)$ is logically equivalent to
 - p
 - $\sim p$
 - q
 - $\sim q$
- The inverse of the proposition $(p \wedge \sim q) \rightarrow r$ is
 - $\sim r \rightarrow (\sim p \vee q)$
 - $r \rightarrow p \wedge \sim q$
 - $(\sim p \vee q) \rightarrow \sim r$
 - None of these
- The negation of $q \vee \sim(p \wedge r)$ is
 - $\sim q \wedge (p \wedge r)$
 - $\sim q \wedge \sim(p \wedge r)$
 - $\sim q \vee (p \wedge r)$
 - None of these
- The contrapositive of $(p \vee q) \rightarrow r$ is
 - $r \rightarrow (p \vee q)$
 - $\sim r \rightarrow (p \vee q)$
 - $\sim r \rightarrow (\sim p \wedge \sim q)$
 - $p \rightarrow (q \vee r)$
- If $p \rightarrow (q \vee r)$ is false, then the truth values of p , q and r are, respectively,
 - T, T, F
 - F, F, F
 - F, T, T
 - T, F, F
- $(p \wedge \sim q) \wedge (\sim p \wedge q)$ is
 - a tautology
 - a contradiction

- (3) neither a tautology nor a contradiction
(4) None of these
21. The proposition $(p \rightarrow \sim p) \wedge (\sim p \rightarrow p)$ is a
(1) tautology and contradiction
(2) neither tautology nor contradiction
(3) contradiction
(4) tautology
22. The false statement among the following is
(1) $p \wedge (\sim p)$ is a contradiction
(2) $(p \rightarrow q) \leftrightarrow (\sim q \rightarrow \sim p)$ is a contradiction
(3) $\sim(\sim p) \leftrightarrow p$ is a tautology
(4) $p \vee (\sim p)$ is a tautology
23. Which of the following is logically equivalent to $\sim(\sim p \rightarrow q)$?
(1) $p \wedge q$ (2) $p \wedge \sim q$
(3) $\sim p \wedge q$ (4) $\sim p \wedge \sim q$
24. If $p \rightarrow (\sim p \vee q)$ is false, the truth values of p and q are, respectively,
(1) F, T (2) F, F (3) T, T (4) T, F
25. The conditional statement $(p \wedge q) \rightarrow p$ is
(1) a tautology
(2) a fallacy
(3) neither tautology nor fallacy
(4) none of these
26. $(p \wedge \sim q) \wedge (\sim p \vee q)$ is
(1) a contradiction (2) a tautology
(3) either (1) or (2) (4) neither (1) nor (2)
27. The proposition $p \rightarrow \sim(p \wedge \sim q)$ is
(1) a contradiction (2) a tautology
(3) either (1) or (2) (4) neither (1) nor (2)
28. $(p \wedge \sim q) \wedge (\sim p \vee q)$ is
(1) a contradiction (2) either (1) or (2)
(3) a tautology (4) neither (1) nor (2)
29. In the truth table for the statement $(p \rightarrow q) \leftrightarrow (\sim p \vee q)$, the last column has the truth value in the following order
(1) TTTT (2) FTFT (3) TTFF (4) FFFF
30. If each of the statements $p \rightarrow \sim q$, $\sim r \rightarrow q$ and p are true, then which of the following is NOT true?
(1) q is false (2) r is true
(3) $r \rightarrow q$ is false (4) $r \wedge \sim q$ is false
31. Which of the following is true?
(1) $p \wedge \sim p \equiv t$ (2) $p \vee \sim p \equiv f$
(3) $p \rightarrow q \equiv q \rightarrow p$ (4) $p \rightarrow q \equiv \sim q \rightarrow \sim p$
32. If p is true and q is false, then which of the following statements is NOT true?
(1) $p \vee q$ (2) $p \wedge (\sim q)$
(3) $p \rightarrow p$ (4) $p \rightarrow q$
33. If $(p \wedge \sim r) \rightarrow (\sim p \vee q)$ is false, then truth values of p , q and r are, respectively,
(1) T, T, T (2) T, F, T
(3) T, F, F (4) F, T, T
34. Statement $(p \rightarrow q) \leftrightarrow (\sim q \rightarrow \sim p)$
(1) is contradiction
(2) is tautology
(3) is neither contradiction nor tautology
(4) none of these
35. The contrapositive of inverse of $p \rightarrow \sim q$ is
(1) $p \rightarrow q$ (2) $\sim q \rightarrow p$
(3) $q \rightarrow p$ (4) $\sim q \rightarrow \sim p$
36. Consider the following statements:
 p : He is intelligent
 q : He is strong
Then symbolic form of statement 'It is wrong that he is intelligent or strong' is
(1) $\sim p \vee \sim q$ (2) $\sim p \vee q$
(3) $\sim p \wedge \sim q$ (4) $p \wedge \sim q$
37. $\sim(p \vee q) \vee (\sim p \wedge q)$ is equivalent to
(1) q (2) p (3) $\sim p$ (4) $\sim q$
38. If $p \rightarrow (q \vee r)$ is false, then the truth values of p , q and r are, respectively,
(1) F, T, T (2) T, T, F
(3) T, F, F (4) F, F, F

Answers Key

EXERCISES

Single Correct Answer Type

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (4) | 2. (1) | 3. (4) | 4. (4) | 5. (1) | 16. (3) | 17. (1) | 18. (3) | 19. (4) | 20. (2) |
| 6. (2) | 7. (3) | 8. (2) | 9. (2) | 10. (4) | 21. (3) | 22. (2) | 23. (4) | 24. (4) | 25. (1) |
| 11. (1) | 12. (1) | 13. (2) | 14. (1) | 15. (2) | 26. (1) | 27. (4) | 28. (1) | 29. (1) | 30. (4) |
| | | | | | 31. (4) | 32. (4) | 33. (3) | 34. (2) | 35. (2) |
| | | | | | 36. (3) | 37. (3) | 38. (3) | | |

Chapter 10

Concept Application Exercises

Exercise 10.1

1.

p	q	$\sim p$	$\sim q$	$\sim p \vee q$	$\sim p \wedge \sim q$	$(\sim p \vee q) \wedge (\sim p \wedge \sim q)$
T	T	F	F	T	F	F
T	F	F	T	F	F	F
F	T	T	F	T	F	F
F	F	T	T	T	T	T

2. (a) $(p \vee \sim r) \wedge (q \vee \sim r)$

p	q	r	$\sim r$	$p \vee \sim r$	$q \vee \sim r$	$(p \vee \sim r) \wedge (q \vee \sim r)$
T	T	T	F	T	T	T
T	T	F	T	T	T	T
T	F	T	F	T	F	F
F	T	T	F	F	T	F
T	F	F	T	T	T	T
F	F	T	F	F	F	F
F	F	F	T	T	T	T

(b) $\sim(p \vee \sim q) \wedge (\sim p \vee r)$

p	q	r	$\sim p$	$\sim q$	$p \vee \sim q$	$\sim(p \vee \sim q)$	$\sim p \vee r$	$\sim(p \vee \sim q) \wedge (\sim p \vee r)$
T	T	T	F	F	T	F	T	F
T	T	F	F	F	T	F	F	F
T	F	T	F	T	T	F	T	F
F	T	T	T	F	F	T	T	T
T	F	F	F	T	T	F	F	F
F	T	F	T	F	F	T	T	T
F	F	T	T	T	T	F	T	F
F	F	F	T	T	T	F	T	F

3. (a) $\sim p \rightarrow (q \rightarrow p)$

p	q	$\sim p$	$q \rightarrow p$	$\sim p \rightarrow (q \rightarrow p)$
T	T	F	T	T
T	F	F	T	T
F	T	T	F	F
F	F	T	T	T

(b) $(p \rightarrow q) \rightarrow (p \wedge q)$

p	q	$p \rightarrow q$	$p \wedge q$	$(p \rightarrow q) \rightarrow (p \wedge q)$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	F
F	F	T	T	T

4. (a) Truth values of $(p \leftrightarrow \sim q) \leftrightarrow (q \rightarrow p)$:

p	q	$\sim q$	$p \leftrightarrow \sim q$	$q \rightarrow p$	$(p \leftrightarrow \sim q) \leftrightarrow (q \rightarrow p)$
T	T	F	F	T	F
T	F	T	T	T	T
F	T	F	T	F	F
F	F	T	F	T	F

(b) Truth values of $(p \rightarrow q) \vee \sim(p \leftrightarrow \sim q)$:

p	q	$\sim q$	$p \rightarrow q$	$p \leftrightarrow \sim q$	$\sim(p \leftrightarrow \sim q)$	$(p \rightarrow q) \vee \sim(p \leftrightarrow \sim q)$
T	T	F	T	F	T	T
T	F	T	F	T	F	F
F	T	F	T	T	F	T
F	F	T	T	F	T	T

5. (a) Truth table for $(p \wedge q) \rightarrow \sim p$

p	q	$p \wedge q$	$\sim p$	$(p \wedge q) \rightarrow \sim p$
T	T	T	F	F
T	F	F	F	T
F	T	F	T	T
F	F	F	T	T

(b) Truth table for $(p \wedge q) \rightarrow (p \vee q)$

p	q	$p \wedge q$	$p \vee q$	$(p \wedge q) \rightarrow (p \vee q)$
T	T	T	T	T
T	F	F	T	T
F	T	F	T	T
F	F	F	F	T

(c) Truth table for $(p \wedge q) \rightarrow r$

p	q	r	$p \wedge q$	$(p \wedge q) \rightarrow r$
T	T	T	T	T
T	T	F	T	F
T	F	T	F	T
T	F	F	F	T
F	T	T	F	T
F	T	F	F	T
F	F	T	F	T
F	F	F	F	T

(d) Truth table for $[p \wedge (\sim r)] \rightarrow (q \vee r)$

p	q	r	$\sim r$	$p \wedge (\sim r)$	$q \vee r$	$[p \wedge (\sim r)] \rightarrow (q \vee r)$
T	T	T	F	F	T	T
T	T	F	T	T	T	T
T	F	T	F	F	T	T
T	F	F	T	T	F	F
F	T	T	F	F	T	T

F	T	F	T	F	T	T
F	F	T	F	F	T	T
F	F	F	T	F	F	T

6. Truth values of $\sim(p \leftrightarrow q)$

p	q	$p \leftrightarrow q$	$\sim(p \leftrightarrow q)$
T	T	T	F
T	F	F	T
F	T	F	T
F	F	T	F

Truth values of $(p \wedge \sim q) \vee (\sim p \wedge q)$

p	q	$\sim p$	$\sim q$	$p \wedge \sim q$	$\sim p \wedge q$	$(p \wedge \sim q) \vee (\sim p \wedge q)$
T	T	F	F	F	F	F
T	F	F	T	T	F	T
F	T	T	F	F	T	T
F	F	T	T	F	F	F

Statements $\sim(p \leftrightarrow q)$ and $(p \wedge \sim q) \vee (\sim p \wedge q)$ have identical truth values.

$$\therefore \sim(p \leftrightarrow q) \equiv (p \wedge \sim q) \vee (\sim p \wedge q).$$

7. Let p be the statement 'Traders do not reduce the prices' and q be the statement 'Government takes action against them'.

The first statement in symbolic form is $p \rightarrow q$ and the second statement is $\sim(p \wedge \sim q)$.

In order to check the equivalence of the above statements, let us prepare the following truth table.

p	q	$\sim q$	$p \wedge \sim q$	$\sim(p \wedge \sim q)$	$p \rightarrow q$
T	T	F	F	T	T
T	F	T	T	F	F
F	T	F	F	T	T
F	F	T	F	T	T

Clearly, $p \rightarrow q$ and $\sim(p \wedge \sim q)$ have same truth values for all the values of p and q . Hence, the two statements are equivalent.

8. **Converse:** If a quadrilateral has two pairs of parallel sides, then it is a rectangle.

Inverse: If a quadrilateral is not a rectangle, then it does not have two pairs of parallel sides.

Contrapositive: If a quadrilateral does not have two pairs of parallel sides, then it is not a rectangle.

Here, both converse and inverse are false. Since the given statement is true, its contrapositive is true.

Exercise 10.2

1.

p	q	$\sim p$	$\sim q$	$p \wedge q$	$(p \wedge q) \vee (\sim p)$	$p \wedge \sim q$	$(p \wedge q) \vee (\sim p) \vee (p \wedge \sim q)$
T	T	F	F	T	T	F	T
T	F	F	T	F	F	T	T
F	T	T	F	F	T	F	T
F	F	T	T	F	T	F	T

For all possible truth values of p and q , the compound statement $(p \wedge q) \vee (\sim p) \vee (p \wedge \sim q)$ is true.

Thus, $(p \wedge q) \vee (\sim p) \vee (p \wedge \sim q)$ is a tautology.

2. Truth values of $[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$.

p	q	r	$p \rightarrow q$	$q \rightarrow r$	$(p \rightarrow q) \wedge (q \rightarrow r)$	$p \rightarrow r$	$[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	T
T	F	T	F	T	F	T	T
F	T	T	T	T	T	T	T
T	F	F	F	T	F	F	T
F	T	F	T	F	F	T	T
F	F	T	T	T	T	T	T
F	F	F	T	T	T	T	T

For all possible truth values of p and q , the compound statement $[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$ is true.

Thus, $[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$ is a tautology.

3. $\sim((\sim p) \wedge q) \equiv \sim(\sim p) \vee \sim q \equiv p \vee (\sim q)$

4. $(p \wedge q) \wedge \sim(p \vee q) \equiv (p \wedge q) \wedge (\sim p \wedge \sim q)$
 $\equiv (p \wedge \sim p) \wedge (q \wedge \sim q)$
 $\equiv f \wedge f$
 $\equiv f$

Thus, $(p \wedge q) \wedge \sim(p \vee q)$ is fallacy, i.e., contradiction.

5. $\sim(\sim p \rightarrow \sim q) \equiv \sim(\sim(\sim p) \vee \sim q)$
 $\equiv \sim(p \vee \sim q)$
 $\equiv \sim p \wedge q$

Exercises

Single Correct Answer Type

1. (4) Clearly, 'please do me a favour' is not a statement.

2. (1) $\sim p$: Ram is not tall

Thus, $\sim p \vee q$ means "Ram is not tall or he is intelligent".

3. (4) All the statements in (1), (2) and (3) are equivalent and each is the negation of p .

4. (4) If p : A number is a prime

q : It is odd

We have $p \rightarrow q$.

The inverse of $p \rightarrow q$ is $\sim p \rightarrow \sim q$.

$\sim p \rightarrow \sim q$: If a number is not a prime then it is not odd.

5. (1) Correct result is: $(\sim p \vee \sim q) \rightarrow (r \wedge s)$

or $\sim(p \wedge q) \rightarrow (r \wedge s)$

6. (2) Let p : Paris is in France, q : London is in England

The given statement is $p \wedge q$.

Its negation is $\sim(p \wedge q) = \sim p \vee \sim q$.

So, the required statement is "Paris is not in France or London is not in England".

7. (3) "Mathematics is interesting" is not a logical sentence. It may be interesting for some persons and may not be interesting for others.

Therefore, this is not a proposition.

8. (2) $\sim((\sim(\sim p)) \wedge q) \equiv \sim(p \wedge q)$

$\equiv \sim p \vee \sim q$

9. (2) $\sim(p \vee (\sim q)) \equiv \sim p \wedge \sim(\sim q) \equiv (\sim p) \wedge q$

$$\begin{aligned}
 10. (4) \quad & \sim(\sim p \rightarrow q) \equiv \sim((\sim(\sim p)) \vee q) \\
 & \equiv \sim(p \vee q) \\
 & \equiv \sim p \wedge \sim q
 \end{aligned}$$

$$\begin{aligned}
 11. (1) \quad & (\sim p \vee q) \wedge \sim r \rightarrow p \\
 & \equiv (F \vee F) \wedge F \rightarrow T \\
 & \equiv F \rightarrow T \\
 & \equiv T
 \end{aligned}$$

$$\begin{aligned}
 12. (1) \quad & (p \wedge \sim r) \rightarrow (q \vee r) \text{ is false.} \\
 & \text{Thus, } p \wedge \sim r \text{ is true and } q \vee r \text{ is false.} \\
 & \text{Hence, } p \text{ must be true.}
 \end{aligned}$$

$$\begin{aligned}
 13. (2) \quad & (p \wedge q) \wedge (q \wedge r) \text{ is true which means that } p \wedge q \text{ and } q \wedge r \text{ are} \\
 & \text{both true.} \\
 & \text{Therefore, } p, q \text{ and } r \text{ are all true.}
 \end{aligned}$$

$$\begin{aligned}
 14. (1) \quad & \sim[p \vee (\sim p \vee q)] \\
 & \equiv \sim p \wedge \sim(\sim p \vee q) \\
 & \equiv \sim p \wedge (\sim(\sim p) \wedge \sim q) \\
 & \equiv \sim p \wedge (p \wedge \sim q)
 \end{aligned}$$

$$\begin{aligned}
 15. (2) \quad & (\sim(p \vee q)) \vee (\sim p \wedge q) \\
 & \equiv (\sim p \wedge \sim q) \vee (\sim p \wedge q) \\
 & \equiv \sim p \wedge (\sim q \vee q) \\
 & \equiv \sim p
 \end{aligned}$$

$$\begin{aligned}
 16. (3) \quad & \text{Inverse of } p \rightarrow q \text{ is } \sim p \rightarrow \sim q \\
 & \text{Therefore, inverse of } (p \wedge \sim q) \rightarrow r \text{ is} \\
 & \sim(p \wedge \sim q) \rightarrow \sim r \\
 & \equiv \sim(p \vee q) \rightarrow \sim r
 \end{aligned}$$

$$\begin{aligned}
 17. (1) \quad & \sim(q \vee \sim(p \wedge r)) \\
 & \equiv \sim(p \wedge (\sim(\sim(p \wedge r))) \\
 & \equiv \sim q \wedge (p \wedge r)
 \end{aligned}$$

$$\begin{aligned}
 18. (3) \quad & \text{Contrapositive of } p \rightarrow q \text{ is } \sim q \rightarrow \sim p \\
 & \text{Therefore, contrapositive of } (p \vee q) \rightarrow r \text{ is} \\
 & \sim r \rightarrow \sim(p \vee q) \\
 & \equiv \sim r \rightarrow (\sim p \wedge \sim q)
 \end{aligned}$$

$$\begin{aligned}
 19. (4) \quad & p \rightarrow q \text{ is false only when } p \text{ is true and } q \text{ is false.} \\
 & \text{Therefore, } p \rightarrow (q \vee r) \text{ is false when } p \text{ is true and } q \vee r \text{ is false.} \\
 & q \vee r \text{ is false only when both } q \text{ and } r \text{ are false.} \\
 & \text{Hence, truth values of } p, q \text{ and } r \text{ are, respectively, T, F and F.}
 \end{aligned}$$

$$\begin{aligned}
 20. (2) \quad & (p \wedge \sim q) \wedge (\sim p \wedge q) \\
 & \equiv (p \wedge \sim p) \wedge (\sim q \wedge q) \\
 & \equiv f \wedge f \\
 & \equiv f \quad (\text{By using associative laws and commutative laws}) \\
 & \text{Therefore, } (p \wedge \sim q) \wedge (\sim p \wedge q) \text{ is a contradiction.}
 \end{aligned}$$

21. (3)

p	$\sim p$	$p \rightarrow \sim p$	$\sim p \rightarrow p$	$(p \rightarrow \sim p) \wedge (\sim p \rightarrow p)$
T	F	F	T	F
F	T	T	F	F

Clearly, $(p \rightarrow \sim p) \wedge (\sim p \rightarrow p)$ is a contradiction.

$$\begin{aligned}
 22. (2) \quad & p \rightarrow q \text{ is logically equivalent to } \sim q \rightarrow \sim p \\
 & \text{Therefore, } (p \rightarrow q) \leftrightarrow (\sim q \rightarrow \sim p) \text{ is a tautology but not a} \\
 & \text{contradiction.}
 \end{aligned}$$

$$\begin{aligned}
 23. (4) \quad & \text{We know that } p \rightarrow q \equiv \sim p \vee q \\
 & \therefore \sim p \rightarrow q \equiv p \vee q \\
 & \therefore \sim(p \rightarrow q) \equiv \sim(p \vee q) \\
 & \equiv \sim p \wedge \sim q
 \end{aligned}$$

$$\begin{aligned}
 24. (4) \quad & p \rightarrow (\sim p \vee q) \text{ is false then } p \text{ is true and } \sim p \vee q \text{ is false.} \\
 & \text{So, } p \text{ is true and both } \sim p \text{ and } q \text{ are false.} \\
 & \text{Therefore, } p \text{ is true and } q \text{ is false.}
 \end{aligned}$$

25. (1)

p	q	$p \wedge q$	$(p \wedge q) \rightarrow p$
T	T	T	T
T	F	F	T
F	T	F	T
F	F	F	T

Therefore, $(p \wedge q) \rightarrow p$ is a tautology.

26. (1)

p	q	$\sim p$	$\sim q$	$p \wedge \sim q$	$\sim p \vee q$	$(p \wedge \sim q) \wedge (\sim p \vee q)$
T	T	F	F	F	T	F
T	F	F	T	T	F	F
F	T	T	F	F	T	F
F	F	T	T	F	T	F

Clearly, $(p \wedge \sim q) \wedge (\sim p \vee q)$ is a contradiction.

27. (4)

p	q	$\sim q$	$p \wedge \sim q$	$\sim(p \wedge \sim q)$	$p \rightarrow \sim(p \wedge \sim q)$
T	T	F	F	T	T
T	F	T	T	F	F
F	T	F	F	T	T
F	F	T	F	T	T

Result is neither tautology nor contradiction.

28. (1)

p	q	$\sim p$	$\sim q$	$p \wedge \sim q$	$\sim p \vee q$	$(p \wedge \sim q) \wedge (\sim p \vee q)$
T	T	F	F	F	T	F
T	F	F	T	T	F	F
F	T	T	F	F	T	F
F	F	T	T	F	T	F

Clearly, $(p \wedge \sim q) \wedge (\sim p \vee q)$ is a contradiction.

29. (1)

p	q	$p \rightarrow q$	$\sim p$	$\sim p \vee q$	$(p \rightarrow q) \leftrightarrow (\sim p \vee q)$
T	T	T	F	T	T
T	F	F	F	F	T
F	T	T	T	T	T
F	F	T	T	T	T

$$\begin{aligned}
 30. (4) \quad & \text{Since } p \text{ is true and } p \rightarrow \sim q \text{ is also true, therefore } \sim q \text{ is true.} \\
 & \text{Also, since } \sim r \rightarrow q \text{ is true, } \sim r \text{ is false and so, } r \text{ is true.} \\
 & \text{From above } r \rightarrow q \text{ is false.}
 \end{aligned}$$

$$\begin{aligned}
 31. (4) \quad & \sim q \rightarrow \sim p \text{ is contrapositive of } p \rightarrow q \text{ and both convey the same} \\
 & \text{meaning.}
 \end{aligned}$$

$$\begin{aligned}
 32. (4) \quad & \text{Since } p \text{ is true and } q \text{ is false, } p \vee q \text{ is true.} \\
 & p \wedge (\sim q) \text{ is true as } \sim q \text{ is true.} \\
 & \text{Also, } p \rightarrow p \text{ is true.} \\
 & \text{Clearly, } p \rightarrow q \text{ is false.}
 \end{aligned}$$

$$\begin{aligned}
 33. (3) \quad & (p \wedge \sim r) \rightarrow (\sim p \vee q) \text{ is false.} \\
 & \text{Thus, } (p \wedge \sim r) \text{ is true and } (\sim p \vee q) \text{ is false.}
 \end{aligned}$$

So, (p is true and $\sim r$ is true) and ($\sim p$ is false and q is false)
Therefore, p is true, r is false and q is false.

34. (2) ($\sim q \rightarrow \sim p$) is contrapositive of ($p \rightarrow q$).

Therefore, ($p \rightarrow q$) $\leftrightarrow$ ($\sim q \rightarrow \sim p$) is tautology.

35. (2) Inverse is $\sim p \rightarrow q$

Contrapositive is $\sim p \rightarrow q$ is $\sim q \rightarrow p$

36. (3) The symbolic meaning of the statement is $\sim(p \vee q) \equiv \sim p \wedge \sim q$.

37. (3) $\sim(p \vee q) \vee (\sim p \wedge q)$

$$\equiv (\sim p \wedge \sim q) \vee (\sim p \wedge q)$$

$$\equiv \sim p \wedge (\sim q \vee q)$$

$$\equiv \sim p \wedge t$$

$$\equiv \sim p$$

38. (3) Proposition $p \rightarrow (q \vee r)$ is false. So, p is true and ($q \vee r$) is false.
Therefore, p is true and q and r are both false.